
Probability and Statistics for Engineers

Reasoning under uncertainty

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A note on names

One long Spanish form of Los Angeles' name

*El Pueblo de Nuestra Señora la Reina
de los Ángeles del Río de Porciúncula*

My full name

José Antonio Aguirre Pérez de León

PhD candidate in Statistics, UC Santa Cruz

Background in applied mathematics and economic theory.

You probably call it

L.A.

You can call me

Antonio or Tony

My research

- Forecasting I develop **deep learning** and Bayesian models for data that change over time.
- Applications Improving environmental and climate forecasts and quantifying uncertainty.
- Software **Statistical software in R** for fitting and evaluating models.

Course essentials

Monday, Wednesday, Friday

1:20–2:25 p.m.

Classroom Unit 002

Ed Discussion Announcements and course questions. Use a **private message** for individual matters.

Canvas Schedule, materials, quizzes, and grades. Open Ed from the course navigation.

Gradescope One readable, handwritten PDF per homework. Assign pages to questions.

Check Ed regularly. Announcements will be posted there.

No Canvas Announcements. Please do not use Canvas Inbox.

Instructor office hours: Friday, 3–4 p.m., GSC Room 204 (in person).

TA discussions and office hours: see Canvas Course support.

Grading at a glance

Assessment	Weight
Homework	25%
Weekly quizzes	5%
In-class activities	5%
Midterm 1 (October 16)	20%
Midterm 2 (November 16)	20%
Final examination	25%

Final-exam arrangements will be announced.

Submit your own reasoning. AI tools may not generate, solve, check, or rewrite assessed work.

One lowest score dropped

Separately in homework, quizzes, and in-class activities. Exams are not dropped.

Attendance and practice

No attendance score. Brief graded activities occur in class without advance notice.

Discussion sections are optional and ungraded.

Deadlines, extensions, and access

All deadlines use Pacific time. Check each posted date.

- | | |
|-----------------------|---|
| Homework | Due at 10 p.m. , with a no-penalty 24-hour grace period . The final cutoff is 10 p.m. the next day. |
| Quizzes | Close at 10 p.m. on the listed date. No grace period. |
| Extensions | No routine extensions. For serious illness or an emergency affecting graded work, contact me privately on Ed promptly. |
| Accommodations | Share your DRC letter privately early. Contact me early about religious observances or access barriers. Approved arrangements are separate from ordinary deadlines and score drops. |

Keep personal or medical details out of public posts. The syllabus gives the full policies.

Probability questions in computing

Google
Search

How likely is a search to take more than two seconds?

Stripe
Payments

If a payment is flagged, how likely is it to be fraudulent?

Netflix
Streaming

How likely is a video to pause for buffering in the next minute?

A smaller average, a larger error

Imagine testing arrival-time predictions for a Waymo ride.

Two software versions, four simulated driving scenarios.

Prediction error in minutes: distance from the actual arrival time. Smaller is better.

	Quiet street	Busy traffic	Rain	Roadworks	Average
Version A	2	2	2	2	2
Version B	0	0	0	6	1.5

Illustrative classroom numbers. Each scenario has equal weight in the average.

Suppose the goal is an error of at most 3 minutes

Which version meets the goal more often in these four scenarios?

Basic definitions

Experiment

A process with possible outcomes we can describe in advance.

Outcome ω

One possible result of the experiment.

Sample space Ω

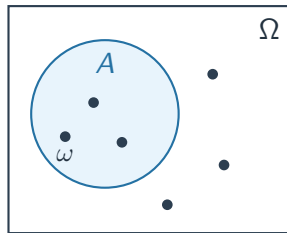
The set of all possible outcomes.

Event $A \subseteq \Omega$

A collection of outcomes that satisfy a description.

An event occurs when the observed outcome belongs to it.

Today we use a finite sample space. Any subset of it is an event.



Each dot is an outcome.
The shaded region is an event.

One experiment: roll a die once

Experiment

Roll a six-sided die once. Record the number on top.

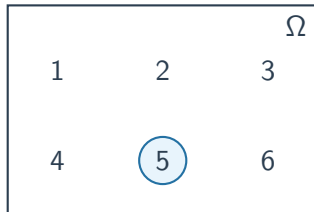
Sample space: all possible results

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

An observed outcome

If the die shows 5, then $\omega = 5$.

One roll produces one outcome from the sample space.



Observed outcome: $\omega = 5$

From a question to an event

What is the probability of rolling at least 5?

In words

Let A be the event “the roll is at least 5.”

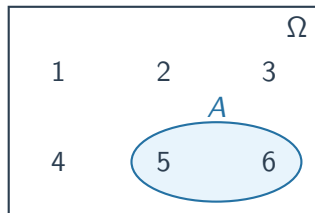
As a set of outcomes

$$A = \{5, 6\} \subseteq \Omega.$$

When does A occur?

When the observed outcome is 5 or 6.

For example, a roll of 5 makes A occur. A roll of 2 makes A not occur.



A simple probability model

For a finite sample space, assign each outcome a probability between 0 and 1. The probabilities must add to 1.

Our assumption: the die is fair.

Each of the six outcomes is equally likely.

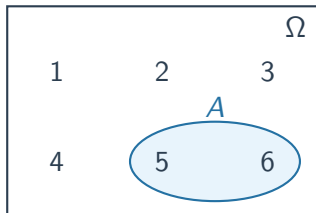
Outcome ω	1	2	3	4	5	6
Probability $P(\{\omega\})$	1/6	1/6	1/6	1/6	1/6	1/6

$$P(\Omega) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1.$$

$P(\{5\})$ means the probability of rolling 5. Fairness justifies the equal probabilities.

The probability of rolling at least 5

Our event is “roll at least 5”: $A = \{5, 6\}$.



Add the probabilities of the outcomes in A .

$$P(A) = P(\{5\}) + P(\{6\}).$$

$$P(A) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \underline{\hspace{1cm}}.$$

$P(A)$ is a number: how likely it is that A occurs.

Each outcome has probability $1/6$.

For a finite sample space with equally likely outcomes

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } \Omega}.$$

The complement: when the event does not occur

What is the probability of rolling less than 5?

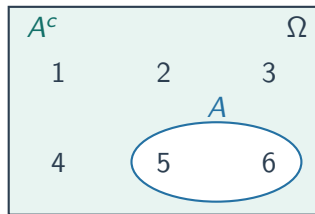
The **complement** A^c means “A does not occur.”

Here, that means “the roll is less than 5.”

$$A^c = \{1, 2, 3, 4\}.$$

$$P(A^c) = \frac{4}{6} = \underline{\hspace{1cm}}.$$

Every outcome is in A or in A^c , and never in both.



The complement rule

$$P(A) + P(A^c) = 1 \quad \implies \quad \boxed{P(A^c) = 1 - P(A).}$$

Here, $1 - 1/3 = 2/3$. The complement rule does not require equally likely outcomes.

Your turn: an even number

Roll the same fair die once.

What is the probability of rolling an even number?

Let B be the event “the roll is even.”

- 1 Write B as a set of outcomes.
- 2 Find $P(B)$.
- 3 Describe B^c in words. Write its outcomes and find $P(B^c)$.

Ω		
1	2	3
4	5	6

Ungraded practice. Compare your reasoning with a neighbor.

Questions we will learn to answer

Question

Probability tools

What can happen?

Events, counting, probability models

What does new information change?

Conditioning and Bayes' theorem

How do numerical outcomes vary?

Distributions, expectation, variance

What do repeated observations tell us?

Limits, estimation, uncertainty

How do random systems evolve?

Poisson processes, Markov chains

The same habit throughout: state the model, reason from it, interpret the result.

Course references

Start with the course material

Use the lecture slides, explanations, and posted notes. Keep your own worked examples.

Supporting reading

Mor Harchol-Balter, *Introduction to Probability for Computing*.

[Author's book page: complete book and selected slides.](#)

Alternative

Morris H. DeGroot and Mark J. Schervish,
Probability and Statistics, fourth edition.

The Weekly Guide lists short selections. You do not need to read both books.

Opening lecture: Harchol-Balter, Sections 2.1–2.2, printed pp. 21–23. Alternative: DeGroot–Schervish, Sections 1.3–1.6.

Before Monday

In Canvas, open **Start Here**.

- ➊ Read the syllabus and check the assessment dates.
- ➋ Open Ed Discussion, enable notifications, and check the announcements.
- ➌ Begin the **ungraded Mathematics Review**. Mark topics where you would like a refresher.
- ➍ Bring your questions. On Monday, we begin with **events and counting**.

First deadlines, both at 10 p.m. Pacific

Quiz 1

October 4

Homework 1

October 6

Homework 1's grace cutoff is October 7 at 10 p.m. Pacific.